Assignment 8: Virtual Memory

Ignore the problem you have done as evaluation problem in the previous week.

1. You are designing a memory management unit (MMU). The system has a fixed number of Physical Frames (e.g., 3 or 4). A process generates a Reference String (a sequence of integers representing page requests). Your job is to determine which pages are in memory at any given time and count the total number of Page Faults.

Logical Clock: Maintain a global integer counter that starts at 0. Every time a page is accessed (whether it's a hit or a fault), increment the counter.

Page Hit: If the requested page is already in a frame, update that frame’s "Last Used Time" to the current counter.

Page Fault (Empty Frame): If the page is not in memory and there is an empty frame (indicated by -1), load the page and set its "Last Used Time" to the current counter.

Page Fault (Eviction): If memory is full, find the frame that has the smallest (oldest) "Last Used Time." Replace that page with the new one and update its "Last Used Time" to the current counter.

2. Implement the 2nd Chance (Clock) algorithm, which is a high-efficiency approximation of LRU used in real kernels like Linux. You have a fixed number of physical memory frames. You are given a stream of page requests (integers).

Each page has a "Reference Bit".

When a page is accessed:

If it's in memory, set its reference bit to 1.

If it's not in memory (Page Fault), find a victim:

Move the clock hand. If the current page's bit is 1, set it to 0 and move to the next.

If the bit is 0, replace this page with the new one and set its bit to 1.

Write a C program that simulates this behaviour. Your program should track the number of Page Faults for a given sequence of page requests.

Use a struct to represent a Frame. It should store the page_id currently held and the reference_bit. Define a fixed number of frames (e.g., MAX_FRAMES).

Maintain a global or static "hand" pointer (an integer index) that persists between function calls.

Your program should read a sequence of page IDs (integers) and output whether each access was a HIT or a FAULT.

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